MARYAM GHADERIAN

GIS & GEOSPATIAL ANALYST

Remote Sensing · Environmental GIS

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PROFESSIONAL PROFILE

GIS and Geospatial Analyst with professional experience in spatial analysis, remote sensing, and environmental mapping. Experienced in integrating and validating satellite, topographic, and field data to produce reliable maps and datasets for technical and environmental studies. Skilled in QGIS and ArcGIS Pro, with recent projects involving Sentinel-1/2 imagery, LiDAR processing, spatial data harmonization, and WebGIS publication. Focused on turning complex geographic data into clear, reliable maps and analysis that support environmental assessment, land-use planning, and practical decision-making.

CORE TECHNICAL SKILLS

GIS & Spatial Analysis: QGIS, ArcGIS Pro, raster/vector geoprocessing, spatial overlay, terrain analysis, land-use and suitability analysis, thematic mapping.

Remote Sensing & Earth Observation: Sentinel-1/2, Landsat, SAR/InSAR, ESA SNAP, SNAPHU, satellite image interpretation, NDVI/NDWI, land-cover analysis, change detection.

Spatial Data Management & QA/QC: Multi-source data integration, spatial data validation, CRS harmonization, georeferencing, metadata review, PostgreSQL/PostGIS.

Geomatics & Field Data: LiDAR and point-cloud processing, GNSS/RTK, UAV photogrammetry, DEM generation, CloudCompare, Agisoft Metashape.

WebGIS & Supporting Tools: GeoServer, WMS/WFS, Leaflet WebGIS, GDAL/OGR, Python-assisted GIS workflows, AutoCAD, HEC-RAS.

PROFESSIONAL EXPERIENCE

GIS Specialist | Part-time / Project-based *Aug 2015 – Aug 2019*

Range and Watershed Management Organization | Golestan Province, Iran

- Integrated topographic, geological, soil, vegetation, land-use, climate, and field GPS data into GIS datasets for environmental and territorial assessments.
- Applied spatial overlay, terrain, and land suitability analysis to support watershed management and natural-resource planning.
- Produced over 45 thematic and planning maps for three management studies covering more than 7,920 hectares, contributing GIS outputs to technical reports.

GIS Analyst | Full-time *Oct 2014 – Oct 2015*

Sabz Eghlim Caspian Company | Gorgan, Iran

- Prepared geospatial layers and thematic maps from satellite imagery for environmental, geotechnical, and land-use studies.
- Integrated GPS/GNSS observations with GIS datasets, checked spatial data for accuracy and consistency, and prepared outputs for technical assessments.

Technical & GIS Analyst | Full-time *Aug 2011 – Aug 2013*

Abnemood Shomal Shargh Consulting Engineers | Golestan Province, Iran

- Processed 1:25,000-scale topographic data in ArcGIS and prepared terrain, drainage, and planning layers for water-resource and infrastructure studies.
- Delineated a 6,755-hectare watershed into 10 sub-basins and standardized spatial datasets for environmental and territorial assessments.

Water Resources & GIS Analyst | Full-time *Aug 2007 – Aug 2009*

Abrah Gostar Tadbir Consulting Engineers | Golestan Province, Iran

- Supported field surveys, river-boundary assessments, and spatial-data validation for four river systems using GIS, HEC-RAS, and AutoCAD.
- Prepared technical maps and geospatial outputs to support water-resource and hydraulic engineering studies.

SELECTED GEOSPATIAL PROJECTS

Orchard WebGIS – MSc Thesis *2025–2026*

Politecnico di Torino | LiDAR, PostGIS, GeoServer, Leaflet

- Processed a 12.3-million-point LiDAR dataset and evaluated four tree-crown delineation methods against 56 reference crowns, achieving a best F1-score of 0.865.
- Harmonized and validated crown geometries and 103 apple observations in PostGIS, published WMS/WFS services through GeoServer, and implemented a local interactive WebGIS prototype.

Sentinel-1 InSAR – Amatrice Earthquake Deformation Mapping *Aug 2026*

Remote Sensing Project | ESA SNAP, SNAPHU, QGIS

- Processed Sentinel-1 radar imagery using interferometric analysis, phase unwrapping, terrain correction, and geocoding to produce a georeferenced ground-deformation map.
- Reviewed coherence and compared the deformation pattern qualitatively with regional earthquake-fault information.

Water & Environmental GeoDecision WebGIS – Turin–Po River *Jul 2026*

Environmental GIS Project | Sentinel-2, QGIS, OpenStreetMap

- Integrated Sentinel-2 NDVI/NDWI, hydrography, water bodies, OpenStreetMap, and Copernicus Urban Atlas data to explore vegetation and water conditions along the Turin–Po River corridor.
- Harmonized spatial datasets in QGIS and presented the results through an interactive environmental WebGIS.

GEO2PLAN – Alpine Geomatics & Climate Adaptation *Aug–Sep 2026*

International Summer School | UAV Photogrammetry, GNSS/RTK, QGIS

- Combined UAV, GNSS/RTK, point-cloud, land-cover, geology, and field data in QGIS to support terrain interpretation, natural-hazard assessment, and territorial planning.

EDUCATION

MSc in Agritech Engineering *2023–2026*

Politecnico di Torino, Italy

MSc in Environmental Engineering *2018–2021*

Gorgan University of Agriculture and Natural Resources, Iran | Environmental Assessment and Territorial Planning

BSc in Water Engineering *2002–2007*

Gorgan University of Agriculture and Natural Resources, Iran

SELECTED TRAINING

NASA ARSET – Fundamentals of Remote Sensing | Esri – Processing SAR Data in ArcGIS Notebooks | FAO – Land-Cover Information for SDG 15

LANGUAGES

English – Full professional proficiency (TOEFL iBT 105) | Italian – A2 | Persian – Native